



Customer Shopping Behaviour Analysis

A data-driven exploration of **3,900 transactions** across product categories — uncovering spending patterns, customer segments, and strategic growth opportunities using Python, PostgreSQL, and Power BI.

Dataset Overview

Dataset at a Glance

3,900 rows · 18 columns

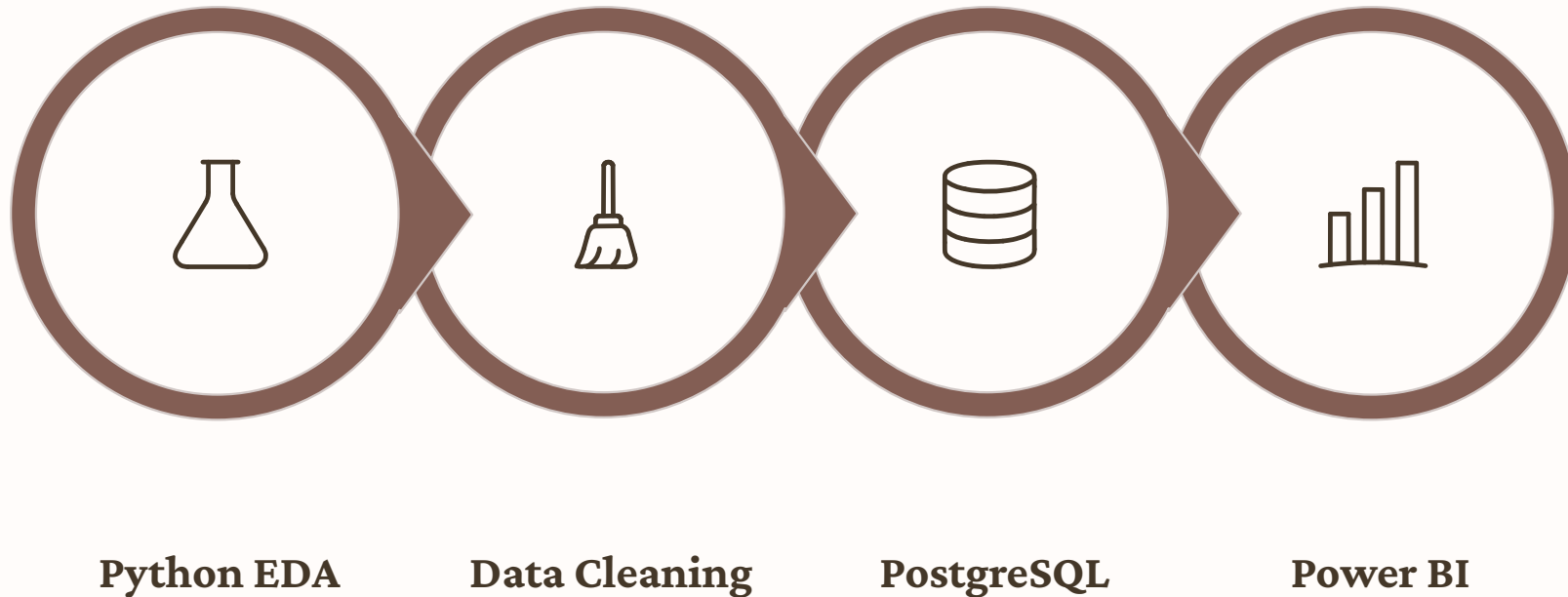
Transactional data spanning demographics, purchases, and shopping behaviour.

📄 37 missing values in Review Rating — imputed using category median.

Key Feature Groups

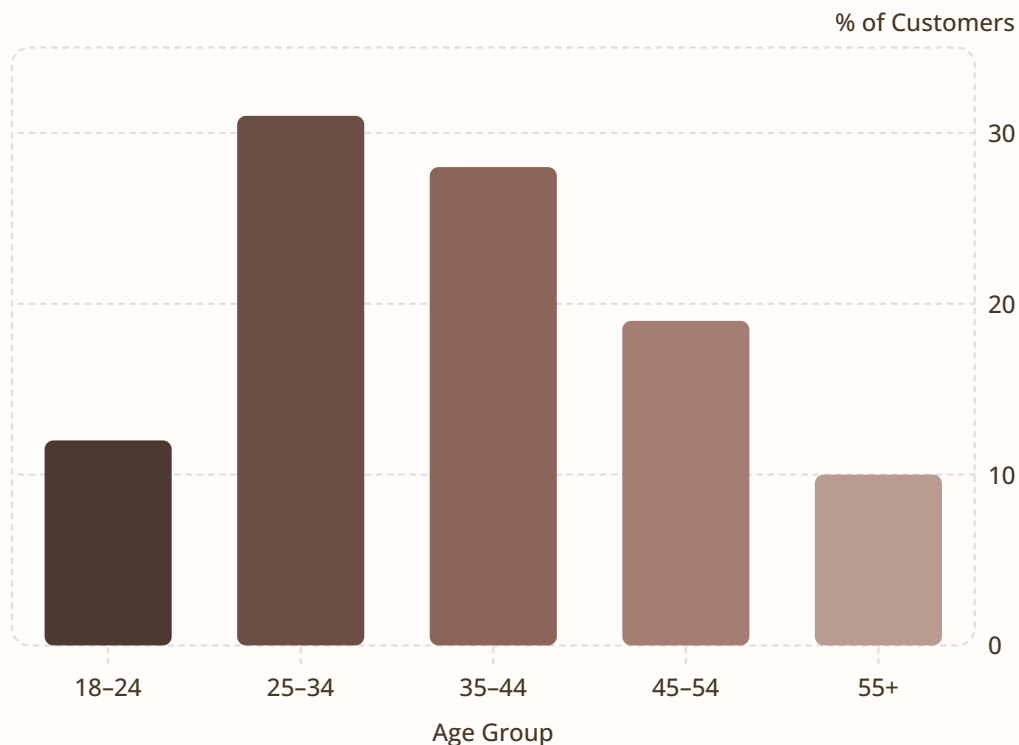
- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Colour
- **Behaviour:** Discount, Previous Purchases, Frequency, Review Rating, Shipping Type

End-to-End Analytical Workflow



The pipeline moved from raw data exploration in Python, through structured SQL analysis in PostgreSQL, to an interactive Power BI dashboard — ensuring reproducible, stakeholder-ready insights at every stage.

Exploratory Data Analysis



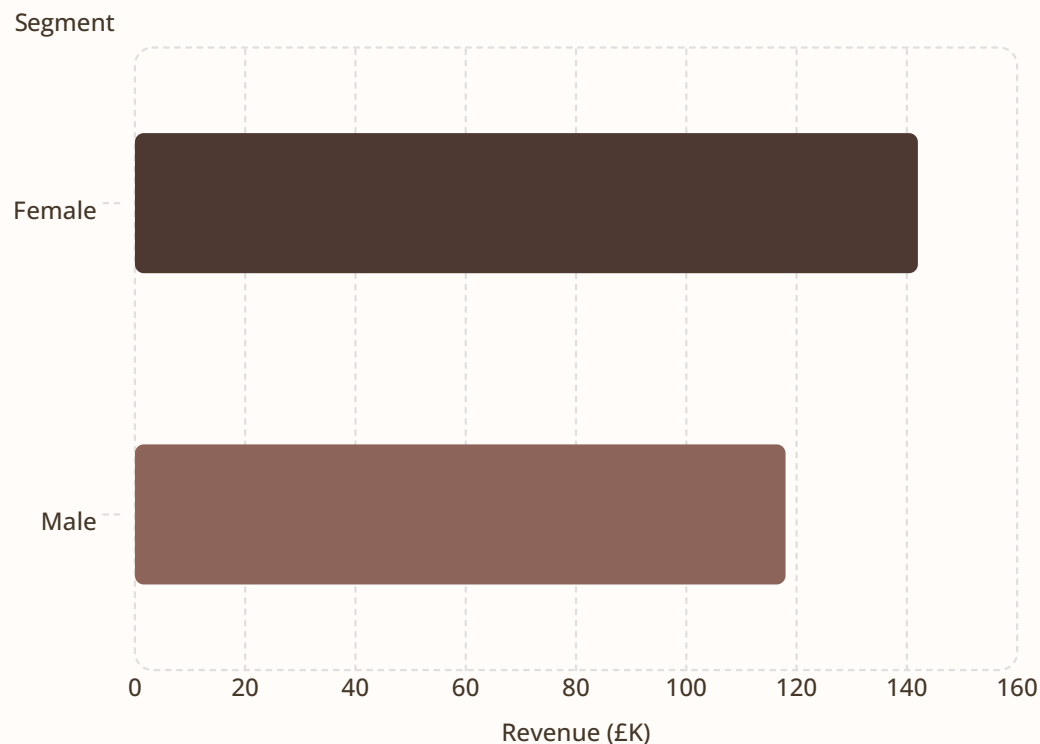
Customer Age Distribution

Customers aged **25-34** form the largest segment (31%), followed closely by 35-44 (28%). The 18-24 and 55+ cohorts represent the smallest shares — a key signal for targeting.

Additional EDA Findings

- Discount applied in **~58%** of transactions
- Review ratings skewed positively — median **4.2 / 5**
- Average purchase frequency: **~4.5 purchases** per customer

Revenue Patterns & Customer Value



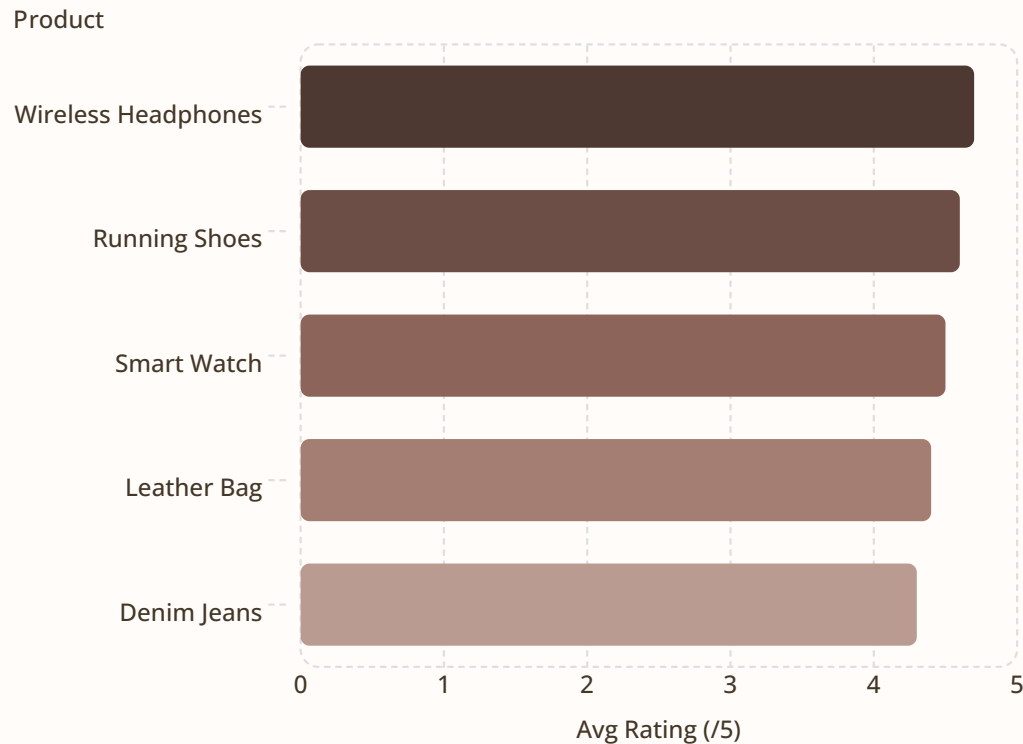
Revenue by Gender

Female customers generated **£142K** vs. **£118K** for males — a 17% difference worth exploring in segmentation.

High-Spending Discount Users

A notable cohort used discounts yet spent **above the average purchase amount** — suggesting discount sensitivity without compromising basket size.

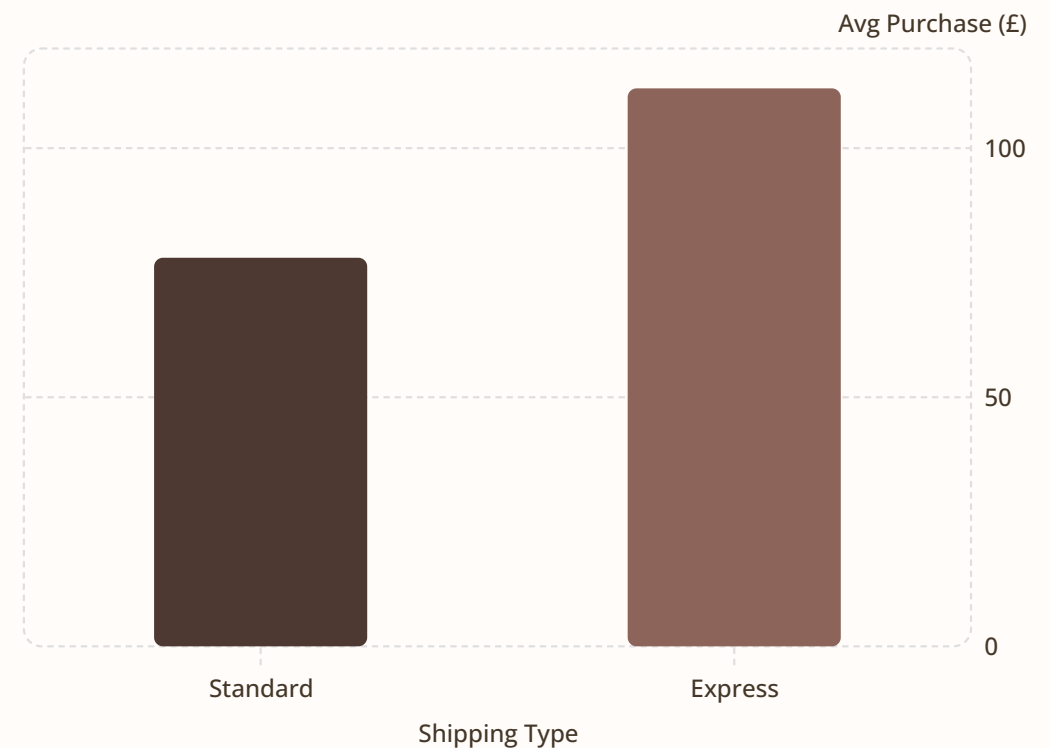
Product Performance & Shipping Behaviour



Top 5 Products by Rating

Electronics and footwear dominate the highest-rated products, all scoring above **4.3/5**.

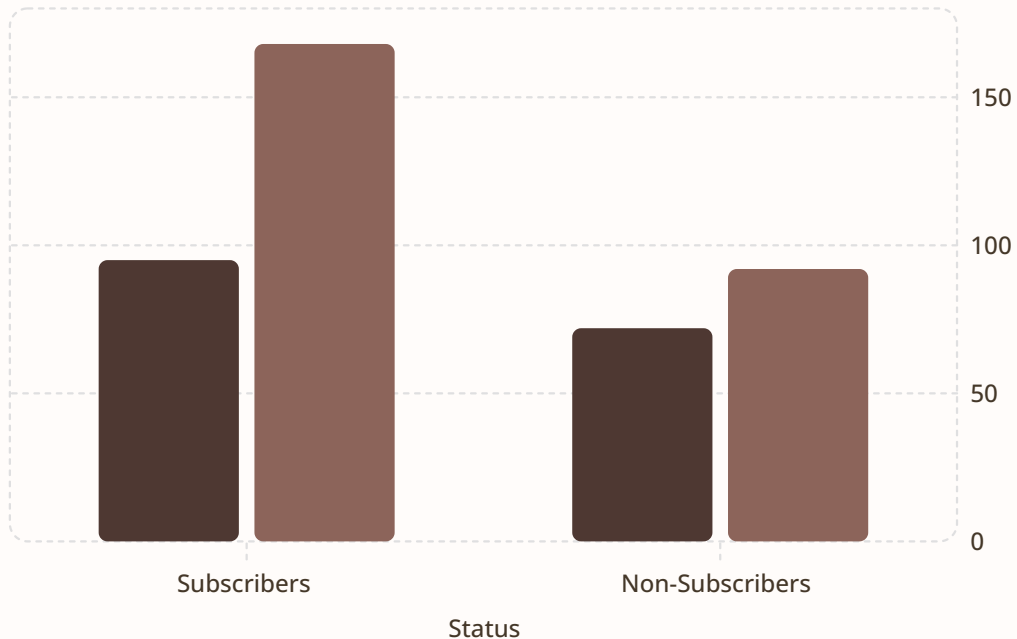
Shipping Type Comparison



Express shipping customers spend **44% more** on average — a strong indicator of higher purchase intent.

Subscriptions, Discounts & Customer Segments

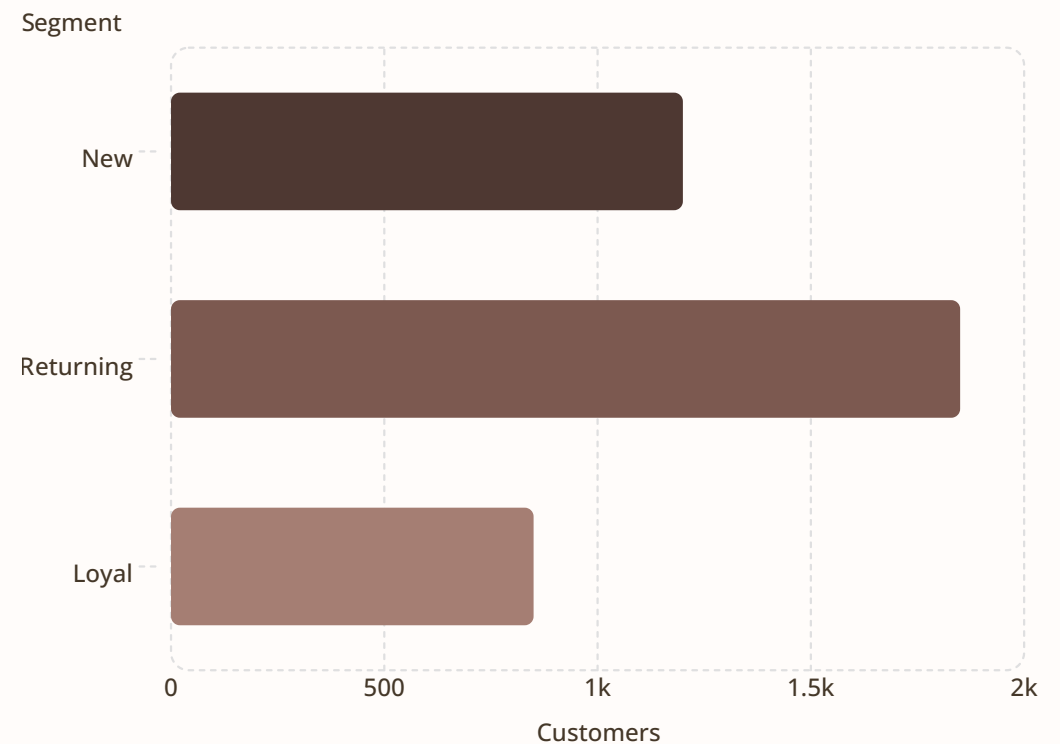
■ Avg Spend (£) ■ Total Revenue (£K)



Subscribers vs. Non-Subscribers

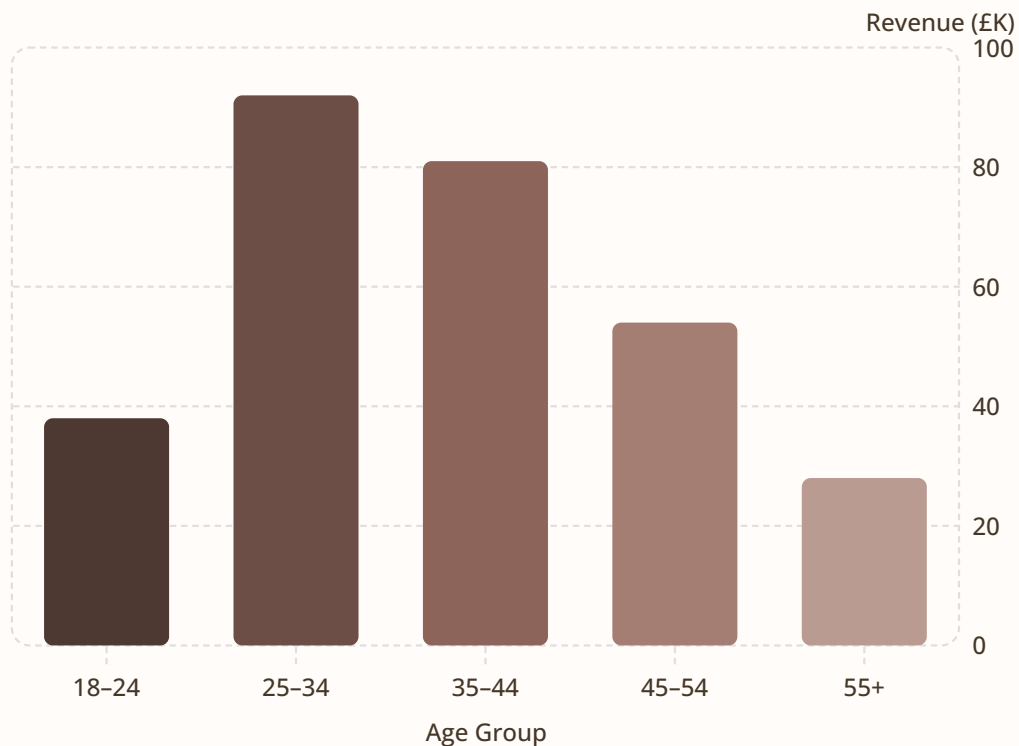
Subscribers spend **32% more** per transaction and contribute **£168K** in total revenue — a compelling case for subscription growth.

Customer Segmentation



Returning customers form the largest group; converting New customers to Loyal is the primary growth lever.

Revenue by Age Group & Repeat Buyer Subscription Rate



Revenue by Age Group

The **25-34** cohort drives the highest revenue (£92K), reinforcing their status as the primary target segment. The 35-44 group is a strong secondary audience.

Repeat Buyers & Subscriptions

Customers with **more than 5 purchases** show significantly higher subscription rates — purchase frequency is a reliable predictor of subscription likelihood and lifetime value.

POWER BI

Interactive Dashboard

An interactive Power BI dashboard was built to present all key insights visually — enabling stakeholders to explore revenue trends, customer segments, product performance, and subscription metrics at a glance.

Revenue Trends

By gender, age group, and shipping type

Customer Segments

New, Returning, and Loyal breakdowns

Product Insights

Top-rated and most-purchased items by category

Subscription Analysis

Spend comparison and repeat buyer conversion

Business Recommendations & Takeaways



Boost Subscriptions

Promote exclusive subscriber benefits — subscribers spend 32% more per transaction.



Loyalty Programmes

Reward repeat buyers to accelerate movement into the "Loyal" segment.



Review Discount Policy

Balance sales uplift with margin control — high-spenders still use discounts.



Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.



Targeted Marketing

Focus on 25–34 and 35–44 age groups, plus express shipping users.

 **Key Takeaway:** These insights directly support strategic decisions on subscriptions, loyalty, pricing, and targeting — driving measurable business growth.